Rank-One Complement Ritz Correction for Dynamic Packet Energy

A Logarithmic-Dimensional Corrector for the RH-88 Dividend

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Abstract

RH-88 showed that old-packet prediction need not contract, whereas variational packet reoptimization restores contraction in all ten archived channels. The open question was whether that reoptimization dividend requires a full ambient-dimensional eigensolve. This paper gives a negative answer at the level of both theorem and validated evidence.

Let $G \geq 0$, let V be an isometric rank-r packet, and choose one unit direction $q \perp V$. In the enriched space $Z = [V, q]$, retain the leading rank-r Ritz subspace of the $(r+1) \times (r+1)$ compression Z^*GZ. The rank-one complement Ritz theorem states that this packet captures at least as much energy as V and gives a valid upper bound on the full optimal tail. Choosing q as the leading left singular direction of $(I - VV^*)GV$ gives maximal cross-block coupling among all single complement directions.

At the final RH-88 update, ranks four through seven require compressed dimensions only five through eight. A 256-bit exact-binary-lift audit certifies that one complement direction reproduces more than 96% of the full floating reference correction dividend in every channel. The corrected tail is at most 3.28 times the full reference tail, and its memory contraction factor remains below 0.24 in all ten channels.

Thus the corrector is logarithmic-dimensional and driven by one cross-block direction at the anchors. A uniform cross-block enrichment bound, Stage A, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: Rayleigh–Ritz; complement enrichment; Ky Fan energy; dynamic packet; cross Gramian; interval arithmetic.

MSC 2020: 47A75; 65F15; 15A18; 65G20.

1 A small corrector for a large ambient problem

RH-88 factored memory contraction into old-packet prediction and a reoptimization dividend [3]. The full corrected packet was obtained from an ambient Gramian. To become an analytic mechanism, the correction must be represented by a small object whose dimension follows the half-logarithmic clock rather than the mesh.

Let G be positive trace class on $\mathcal{K}$, let $V : \mathbb{C}^r \to \mathcal{K}$ be an isometry, and write $P = VV^*$. Choose a unit vector $q \in \operatorname{ran}(I - P)$ and define $Z = [V, q] : \mathbb{C}^{r+1} \to \mathcal{K}$.

2 Rank-one complement Ritz theorem

Theorem 2.1 (Rank-one complement Ritz theorem)**.** *Let $C : \mathbb{C}^r \to \mathbb{C}^{r+1}$ contain the leading r orthonormal eigenvectors of $H = Z^*GZ$, and put $U = ZC$. Then*

$$\operatorname{tr}(U^*GU) = \sum_{k=1}^{r} \lambda_k(H) \geq \operatorname{tr}(V^*GV). \tag{1}$$

Consequently

$$E_r(G) \le \operatorname{tr} G - \operatorname{tr}(U^*GU) \le \operatorname{tr} G - \operatorname{tr}(V^*GV). \tag{2}$$

The corrected packet requires only an $(r+1)$-dimensional eigenproblem.

Proof. The old packet range is an admissible rank-r subspace of $\operatorname{ran} Z$. Ky Fan's maximum principle applied to the compressed Hermitian matrix H gives (1). The full optimal rank-r captured energy is at least the restricted Ritz energy, yielding (2) [1, 2]. □

The theorem controls captured energy without comparing principal angles or small tail gaps. It also explains why a one-pair swap can be too weak: the Ritz solve may rotate all r retained directions jointly after adding only one new complement direction.

Proposition 2.2 (Maximal cross-block coupling)**.** *Let $Q = I - P$ and $B = QGV$. Among unit $q \in \operatorname{ran} Q$,*

$$\max_q \|q^*GV\|_2 = \|B\|,$$

and any leading left singular vector of B attains the maximum.

Proof. The objective is $\|q^*B\|_2$. Its maximum over unit left vectors is the largest singular value of B. □

This proposition selects a canonical one-dimensional correction from the old-packet/new-Gram cross block.

3 Ten-channel 256-bit audit

At the last RH-88 predictor-corrector update, we form the normalized memory Gramian, the old clock-rank packet, and the leading cross-block direction. The corrected packet is obtained from a compressed matrix of dimension $r + 1 \le 8$. All frozen binary Gramians and packet entries are lifted exactly to 256-bit Arb arithmetic. Candidate residual energies are evaluated in the form

$$\operatorname{tr} G - 2\operatorname{tr}(U^*GU) + \operatorname{tr}\big((U^*U)(U^*GU)\big),$$

which remains a valid rank-r residual even under tiny lifted orthogonality defects.

σ	Ritz dimension	min dividend fraction	max tail ratio	max contraction
0.16	5	0.9965	1.481	0.0043
0.08	6	0.9788	3.273	0.0212
0.04	7	0.9612	1.500	0.0584
0.02	7	0.9810	1.034	0.2385
0.01	8	0.9973	1.014	0.1900

Table 1: Worst directional channel at each scale, rounded conservatively. The dividend reference is the full binary64 packet at the same update.

The minimum fraction occurs at $\sigma = 0.04$ and still exceeds 96%. The largest corrected/reference tail ratio occurs at $\sigma = 0.08$, where the absolute contraction remains below one percent. At the hardest contraction scale $\sigma = 0.02$, the small Ritz corrector remains below 0.24.

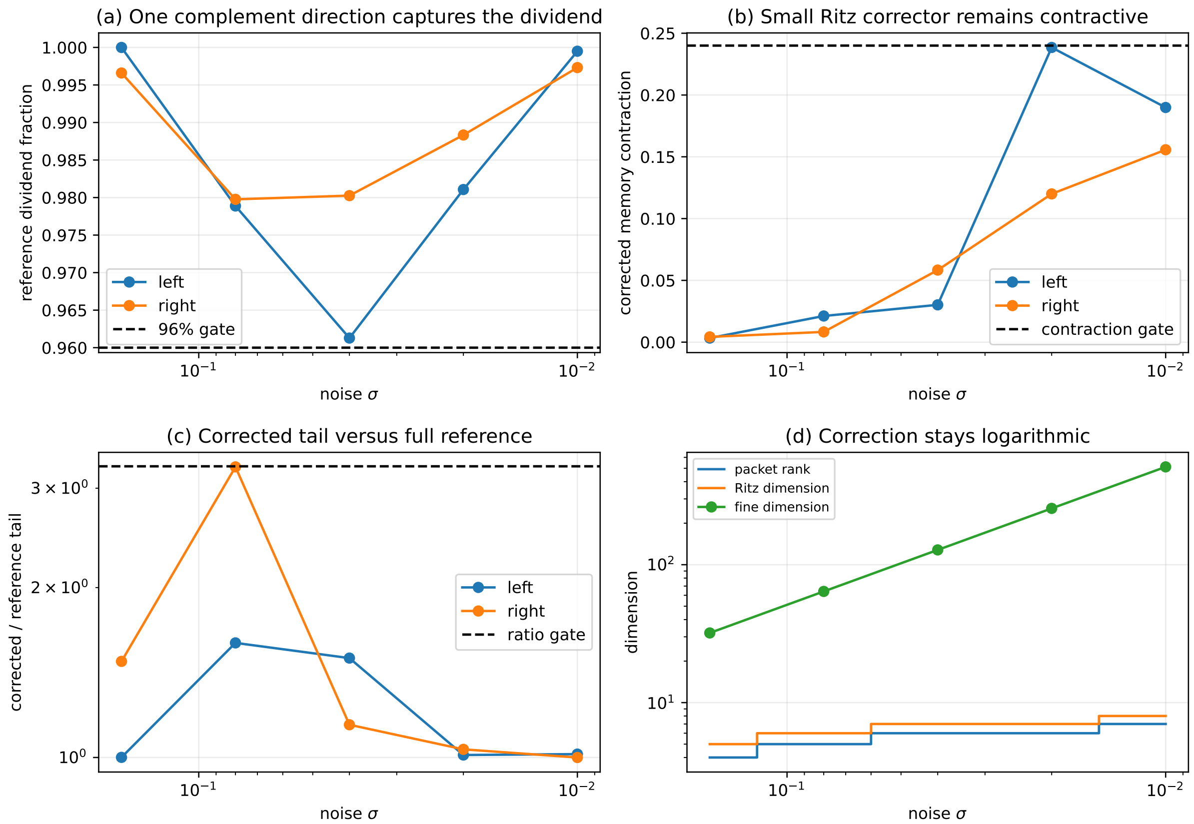

Figure 1: Reference dividend fraction, corrected contraction, corrected tail ratio, and compressed dimension versus the ambient mesh.

4 Boundary and next theorem

The all-level route now has a concrete corrector criterion. It is sufficient to prove that the leading cross-block complement direction yields a fixed fraction of the old-packet Ky Fan deficit, while the resulting corrected factor stays below one after burn-in. Such a bound may be attacked through a Schur complement of the (P, Q) block Gramian rather than full spectral perturbation theory.

RH-89 proves the Ritz and maximal-coupling statements and validates a small corrected packet at ten anchors. It does not prove uniform cross-block enrichment, close Stage A1 or Stage A4, construct a relative determinant or a self-adjoint Hilbert–Polya operator, identify zeta zeros, or prove the Riemann Hypothesis.

References

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